Automatic Spectral Representation With Improved Stacked Spectral Feature Space Patch (ISSFSP) for CNN-Based Hyperspectral Image Classification

Hui Chen, Dameng Yin, Jin Chen, and Yang Chen

***Abstract*— Hyperspectral imaging (HSI) effectively supports land cover (LC) classification in understanding nature–human interactions. The LC classification quality depends heavily on the input features. Combining convolutional neural network (CNN) and stacked spectral feature space patch (SSFSP) has proved effective in simultaneously exploiting spatial and spectral features for accurate LC classification. SSFSP is a stack of gridded spectral feature space images (GSIs) where the spatial and intensity distributions of LCs are derived and directly learned by CNN. However, the GSI set was not fully optimized for HSI classification. In this article, we propose an improved SSFSP (ISSFSP) to tackle this problem, by integrating SSFSP with a novel feature selection (FS) method called band combination search (BCS). BCS first defines a novel metric to measure the ability of a GSI to distinguish various LCs, and then uses a coarse-to-fine strategy to efficiently identify the optimal GSI set. Furthermore, BCS provides a classifier-free method for determining the appropriate number of GSIs after trading off accuracy and efficiency. We conducted experiments using three HSI datasets, ISSFSP and five state-of-the-art FS methods, and six classification models to validate the effectiveness of ISSFSP. ISSFSP obtained consistently highest accuracy (over 98% overall accuracy, 97% average accuracy, and 97% Kappa) using fewer bands and samples, indicating that ISSFSP significantly boosts the performance of CNN while reducing the dependence on model structure and sample quality. In short, ISSFSP can automatically provide promising spectral features for CNN-based HSI classification regarding classification accuracy and model independency, showing great potential for practical applications.**

***Index Terms*— Convolutional neural networks (CNNs), handcrafted feature, hyperspectral imaging (HSI) classification, improved stacked spectral feature space patch (ISSFSP), land cover (LC) classification.**

I. Introduction

THE continuous development of imaging equipment in terms of spatial and spectral resolution and miniaturization has contributed to the popularity of hyperspectral imaging (HSI) technology, which is available from satellites, aircraft, and now unmanned aerial vehicles (UAVs). HSI acquires the spectral response of terrestrial targets in narrow bands of dozens to hundreds, can provide far more informative spectral profiles of terrestrial targets than multispectral imaging (MSI), and is thus applied to many applications, such as target detection, precision farming, pest detection, and land cover (LC) classification [1], [2], [3]. LC classification is one of the most widely used scenarios for HSI. However, the increased dimensionality also brings about nonnegligible computational complexity and high redundancy of information among adjacent bands [4], as well as the curse of dimensionality [5]. Dimensionality reduction (DR) is often required in HSI classification to tackle these above-mentioned issues. The idea of DR is to select or generate the most informative set of bands or new features from HSI. DR can be categorized into feature extraction (FE) and feature selection (FS). FE combines band sets to generate completely new features, while FS picks out raw spectral bands with desired numbers based on criteria, such as similarity and information entropy [6], [7], [8], [9]. FS has been deemed preferable in many studies because of the high interpretability and easy transferability of the selected features [7], [10].

The quality of the extracted or selected features plays a decisive role in an HSI LC classification system. Therefore, much effort has been devoted to designing features for better representations of LCs [11]. Apart from directly using the informative features obtained by DR for classification, further feature mining is usually performed on these features. According to the generation process, the features can be generally categorized into conventional handcrafted features and the emerging deep features in recent years [12]. Handcrafted features are selected or designed based on prior knowledge or statistical characteristics to enhance the spectral and spatial-texture separability of LCs. Therefore, handcrafted features have solid mathematical foundations, a strong correlation with LC properties, and simple training requirements; however, it is challenging to extract comprehensive handcrafted features with ever-increasing data types, volume, and multimodality [13], [14].

Deep features are generally derived by deep learning (DL) [15]. For LC classification, convolutional neural networks (CNNs) have gained increasing attention for their power to automatically extract domain-oriented high-level discriminative features from input data [16]. While CNNs

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can achieve superior accuracy, constructing, fine-tuning, and using an accurate CNN is time-consuming and knowledge-dependent. A great deal of work has been devoted to updating the structure of CNNs to better mine spatial and spectral features [15]. It has been demonstrated that using handcrafted spatial features as CNN input can achieve promising classification accuracy without a complex model [17], [18], [19]. The handcrafted features used in these works have, however, predominantly focused on the spatial aspect. To fully exploit the rich spectral information contained in HSI, methods to automatically handcraft spectral features are necessitated.

Recently, Chen et al. [20] proposed a stacked spectral feature space patch (SSFSP) for crop classification, which paved a new path to combine handcrafted spectral features with CNN. SSFSP is a stack of gridded spectral feature space images (GSIs) that were derived by gridding multiple two-dimensional feature spaces (2DFSs). Each 2DFS is composed of a two-band combination (TBC). With the gridding transformation, the spatial distributions and intensity distributions of various LCs in a 2DFS are recorded in its corresponding GSI. Therefore, the implicit spectral relationships in an HSI are represented as an explicit spatial feature. Through the combination of the SSFSP and CNN, both high-level spectral and spatial features can be mined simultaneously; however, the SSFSP can still be improved. The TBCs used to generate GSIs are selected with reference to the central wavelengths of Landsat-8 bands; thus, the selected TBCs may not be optimal for SSFSP because the ability of the HSI narrow band to characterize the ground surface is not fully utilized. In addition, it may suffer from introducing less informative bands leading to an excessive data volume. Furthermore, it also lacks guidance on the prioritization of TBCs and how many TBCs to be selected when trading off classification accuracy and efficiency. With the above recognition, the existing FS methods may not be applicable to SSFSP for practical HSI LC classification.

We propose improved SSFSP (ISSFSP) to cope with the above limitations. The ISSFSP is the combination of SSFSP and an FS algorithm called band combination search (BCS). The innovations of ISSFSP are as follows.

1) Proposing a metric to define the quality of a GSI by measuring the separability of various LCs in a GSI.
2) Adopting a coarse-to-fine strategy to accelerate the process of optimal TBC searching.
3) Presenting a classifier-free guidance on selecting an appropriate number of TBCs.

ISSFSP is a modification of SSFSP that can rank possible TBCs according to the dataset and automatically select the optimal number of TBCs in a classifier-independent manner, then generate advanced spectral representations from an HSI for CNN-based classification. The remainder of this article is structured as follows. In Section II, we present the related work of our proposed approach. In Section III, we give a thorough description of ISSFSP. The experimental settings and experimental results are presented in Section IV, followed by the discussion and conclusion in Sections V and VI, respectively.

II. Related Work

In this section, we describe three aspects that are relevant to our work, namely, FS, CNN-based HSI classification, and handcrafted features for CNN-based HSI classification.

A. Feature Selection

Depending on whether label information is used, FS methods can be classified as unsupervised FS and supervised FS.

Without class information, unsupervised FS needs to define metrics to measure the quality of bands for the classification task. For example, the product of local spectral density and information entropy is used in the fast neighborhood grouping method for hyperspectral band selection (FNGBS) [9]. Structural similarity (SSIM) is used to choose a set of bands that have the highest global similarity to other bands [8]. Fast density-peak-based clustering (FDPC) assigns a higher ranking to a band that has high density and is at a large distance from other bands that have high density [21]. Recently, a method has been proposed to combine position embedding and importance aggregation modules for raw bands weighting (BW). The position embedding module models remote dependencies in one spatial direction while retaining accurate position information in the other, whereas the importance aggregation module aggregates the global information [22].A common limitation of these individual band ranking methods is that they are, however, sensitive to noise and may select bands with high similarity [7]. Another kind of unsupervised FS, grouping-based methods, first divide bands into groups considering that adjacent bands have high similarity to each other and then select a representative band from each group [6]. Typical approaches for defining the groups include hierarchical clustering [23], spectral clustering [24], and affinity propagation [25]. In addition, strategies such as dynamic planning [6] and image linear reconstruction [7] are also used for band grouping. The bands selected by the grouping-based methods, however, may not have much information for classification.

When labels are available, supervised FS designs specialized objective functions oriented to class information or be based on classifiers to aid in band selection. For example, a high dimensional model representation (HDMR) uses relatively low-order correlations of the input variables to simplify a high dimensional model, and a k-nearest neighbor (KNN) classifier is used to rank the importance of features [26]. In general, the use of classifiers to assist in FS is a time-consuming method. Furthermore, supervised FS does not always give better results than the unsupervised FS [27].

Both unsupervised and supervised FS select individual bands. As the SSFSP uses multiple TBCs as input, the existing FS methods may not be optimal.

B. CNN-Based HSI Classification

In the last decade, CNNs with different structures have been proposed to extract features and thus improve mapping accuracy. In the view of kernel shape, CNNs with a one-dimensional (1-D) kernel (1-D-CNN) take the spectral vector of individual pixels as input and can extract spectral features

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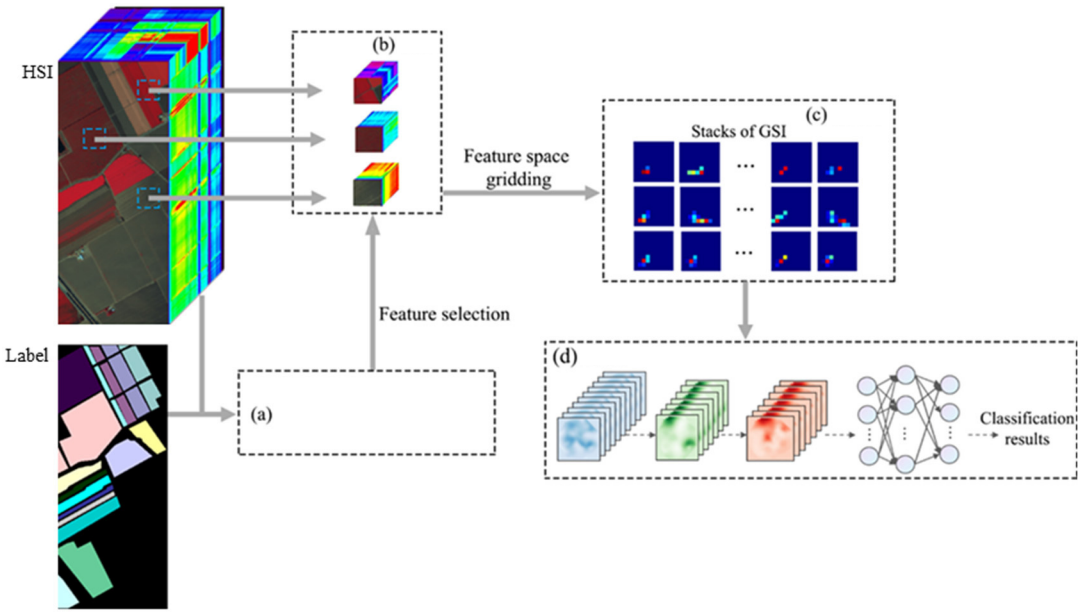


Fig. 1. Schematic of using the ISSFSP for CNN-based HSI classification. The ISSFSP is the combination of SSFSP and BCS. (a) BCS first automatically selects an optimal set of TBCs based on the labels of the training samples. (b) Subsequently, each pixel to be classified is cropped in a regular shape geometry (we call it TFP). (c) TFPs are then transformed into SSFSPs guided by the selected TBCs and (d) then fed into the CNN for training.

well [28], [29] but do not take spatial information into account. The 2-D-CNNs perform convolutional operations in spatial dimensions, which can well consider the texture and spatial relationship of terrestrial targets [30], [31], but their drawback in insufficient spectral-feature mining has also been recognized. The 3-D-CNN performs simultaneous convolutional operations in spatial and spectral dimensions [32]. However, the large number of parameters to train significantly increases the computational burden and the possibility of model overfitting [32]. Attempts have been made to combine their respective strengths. For example, using a dual-channel CNN structure, the spectral and spatial features are extracted in parallel with 1-D-CNN and 2-D-CNN [33]. In [34], a hybrid CNN is composed of a spectral–spatial 3-D-CNN and subsequently a spatial 2-D-CNN, thereby significantly reducing the complexity of the model compared to using two 3-D-CNNs. Since terrestrial targets vary in geometry, there are also models that employ a multibranch fusion strategy to extract features by using multisized convolutional kernels [35] or by selecting a suitable CNN based on the size of the terrestrial target [36]. Attention mechanisms have also been successfully applied to the spectral level [37], spatial level [38], and feature-spatial level [39]. Although these models have made progress in classification performance, their sophisticated structures aggravate the difficulty of their training and optimization and impose higher requirements on the quality and quantity of samples.

C. Handcrafted Features for CNN-Based HSI Classification

The strategy of combining handcrafted features and CNN is common for CNN-based HSI classification. For example, many frameworks first conduct principal component analysis (PCA) on the raw spectral data, and then feed the top components into CNNs [40]. Similarly, maximum noise fraction (MNF) is widely used [41] to reduce the dimensionality of input data and for feature representation. Handcrafted features can achieve the same purpose. For example, Gabor features are used as input to the CNNs [42], [43] or Gabor features and CNN features are extracted separately and then fused [35]. Other handcrafted features such as local binary pattern (LBP) [19], gray-level cooccurrence matrix (GLCM) [18], and extended morphological profiles (EMPs) [44] have also been used as the input of CNNs for HSI classification. The performance of these methods is improved compared to the original CNNs, but these handcrafted features do not leverage well the advantages of CNN mining spatial patterns to extract the rich spectral information contained in the HSI.

III. Methodology

In this section, we give a thorough introduction to the ISSFSP, including a brief review of SSFSP, the steps of generating ISSFSP, and details of two major improvements in ISSFSP, a new quality index (QI) for TBC and a new procedure of BCS. The schema of using ISSFSP for CNN-based HSI classification is presented in Fig. 1.

A. Brief Review of SSFSP

In an HSI, although intraclass differences exist, pixels belonging to the same LC types are clustered in the feature space, while pixels of different LCs have their own clustering centers. This phenomenon is demonstrated in Fig. 2(a), where the distributions of five LCs in multiple 2DFSs generated with multiple TBCs are exhibited. Clearly, every 2DFS provides the spatial distribution information and intensity distribution information of LCs. Meanwhile, 2DFS is a 2-D space, which inspired us to grid it into a GSI [Fig. 2(b)] so that it can be

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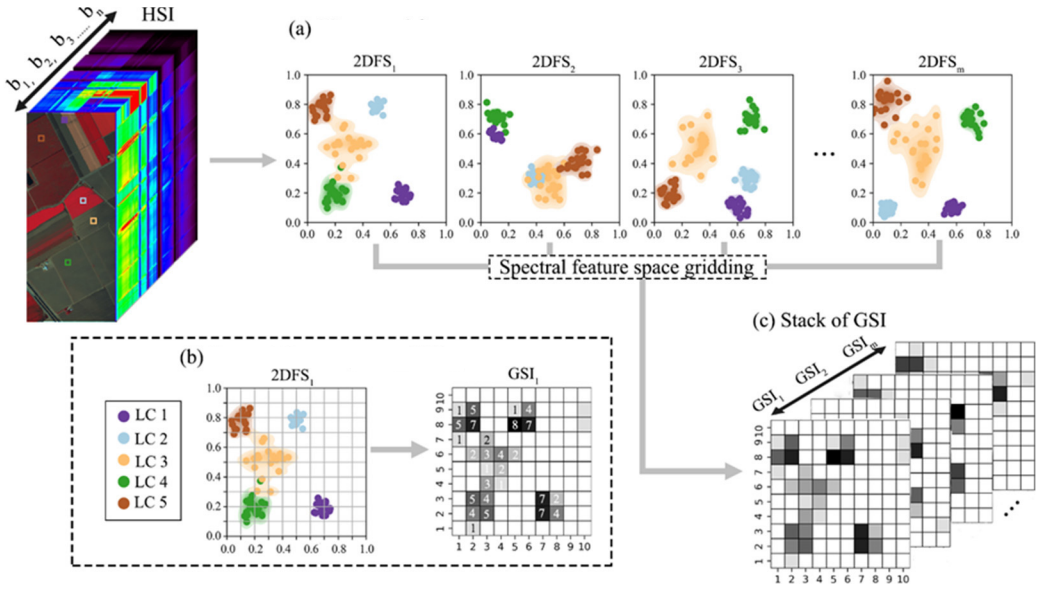


Fig. 2. Scheme for generating a stack of GSIs. (a) Each 2DFS is generated with two HSI bands. (b) Spatial distribution and intensity distribution in each 2DFS are recorded via spectral feature space gridding. (c) Combining these GSIs in a stacked form can make full use of the ability of CNNs for spatial feature mining.

directly processed by CNN. Through such an operation, the implicit spectral relationship contained in multiple TBCs is presented explicitly in the form of spatial features by a stack of GSIs [Fig. 2(c)]; thus, the powerful spatial feature mining capability of CNN can be utilized. However, for an HSI with hundreds of bands, the number of possible TBCs is enormous. To process this many TBCs is computationally intense and time-consuming. Hence, it is necessary to select a subset of TBCs with the strongest differentiability for the target LCs, simultaneously considering the accuracy and efficiency.

B. Steps of Generating ISSFSP

In a normal CNN-based HSI classification framework, a pixel is classified according to the spatial features contained in a regular-shaped patch surrounding it. The patch, which is a stack of spectral bands, is referred to as the traditional feature patch (TFP) in this study [Fig. 1(b)]. Inspired by traditional CNNs, the original SSFSP converts TFP into multiple GSIs, aiming to amplify the spectral information in the feature space and thus facilitate the classification of central pixels [Fig. 1(c)]. This new representation of the TFP, i.e., a stack of GSIs, which is now guided by the BCS, is referred to as the ISSFSP. The process of generating an ISSFSP from its corresponding TFP involves the following three steps.

1) *Selecting TBCs:* As previously mentioned, the number of possible TBCs is huge for an HSI, and TBCs need to be filtered in advance by BCS. For ease of reading, BCS is presented in Sections III-C and III-D. For an HSI $H \in \mathbb{R}^{m\times n\times c}$, multiple TBCs are formed by BCS, and H is reduced to its subset image $H' \in \mathbb{R}^{m\times n\times c'}$, where

$$\mathrm{TBC}_{i,j} = \left(b_i, b_j\right), \quad b_i, b_j \in H,\ i < j \tag{1}$$

m and n are the spatial size (height and width), c and c' are the spectral band number of H and H' ($c \ll c'$). b_i and b_j are the ith and jth bands of H.

2) *Generating TFP:* To make spectral values comparable between bands, H' is standardized by the global maximum (max) and the minimum (min)

$$H'_{\mathrm{stand}} = \frac{H' - \min}{\max - \min}. \tag{2}$$

Then, $\mathrm{TFP} \in \mathbb{R}^{w\times w\times c'}$ is cropped from H'_{stand}, where w is a user-defined parameter to specify the window size of neighboring pixels.

3) *Generating ISSFSP:* Every GSI of ISSFSP is generated under the guidance of a TBC. For $\mathrm{TBC}_{i,j}$, the coordinate of the kth pixel of a TFP in $\mathrm{GSI}_{i,j}$ is

$$\mathrm{coord}^{\mathrm{GSI}}_{i,j}(k) = \mathrm{Round}\left(R\left(I^{\mathrm{TFP}}_i(k), I^{\mathrm{TFP}}_j(k)\right)\right) \tag{3}$$

where $I^{\mathrm{TFP}}_i(k)$ and $I^{\mathrm{TFP}}_j(k)$ are the standardized value of the kth pixel of TFP at bands i and j, which compose $\mathrm{TBC}_{i,j}$, R is a scaling factor that decides the coordinate ranges in each GSI (for example, 10 in Fig. 2). Round() is the rounding function to convert decimal numbers into their nearest integers. After that, the number of pixels of TFP that have the same coordinate in $\mathrm{GSI}_{i,j}$ is summarized and assigned to the value of the corresponding grids (Fig. 2). All the GSIs constitute ISSFSP $\in \mathbb{R}^{R\times R\times c^{\mathrm{TBC}}}$, where c^{TBC} is the number of TBCs selected by BCS.

C. Index for Measuring the Quality of a TBC

Before we present the BCS, we first give the QI to measure the quality of TBSs with the aid of labeled samples from the following three aspects.

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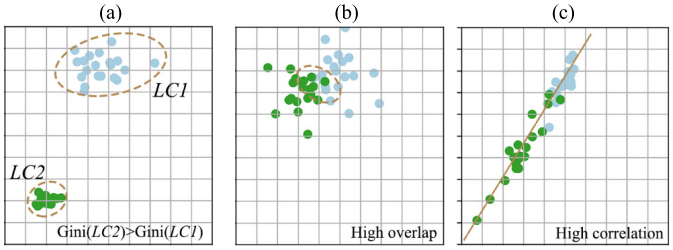


Fig. 3. Three considerations for choosing optimal TBCs. In a 2DFS composed by a TBC, LCs are expected to have (a) high spatial aggregation, (b) low degree of overlap in spatial terms, and (c) low colinearity.

1) *Spatial Aggregation:* An LC with a wide spatial distribution in the GSI is not expected because it implies a high intraclass variance, i.e., potentially low separability, in the corresponding two bands [Fig. 3(a)]. We introduce the spatial Gini index [45] to measure the spatial aggregation for LCs under $\mathrm{TBC}_{i,j}$

$$\mathrm{Gini}_{i,j} = \sum_{l=1}^{L} \sum_{k=1}^{R\times R} \left(S_{l,k}^{i,j} - Z_k^{i,j} \right)^2 \tag{4}$$

where $S_{l,k}^{i,j}$ and $Z_k^{i,j}$ are the proportion of samples of the lth LC and all LCs in the kth pixel of $\mathrm{GSI}_{i,j}$, and L is the number of LCs.

2) *Spatial Overlap:* Two LC types with a high degree of overlap indicate that the interclass variance between them is small [Fig. 3(b)]. Thus, we calculate the proportion of samples of all LCs falling in the overlapping areas

$$\mathrm{OL}_{i,j} = \frac{N^{\mathrm{ol}}}{\sum_{l=1}^{L} N_l^{\mathrm{ol}}} \tag{5}$$

where N_l^{ol} and N^{ol} are the sample number of the lth LC and of all LCs that fall in the overlapping areas.

3) *Correlation Coefficient:* Apart from the above two indices, it is also common for samples to have a high correlation on some bands [Fig. 3(c)]. Such a TBC needs to be excluded because the information provided by these two bands is redundant. We use the correlation coefficient to measure the collinearity of all LCs in a TBC

$$\mathrm{Cor}_{i,j} = \frac{\mathrm{Cov}(\boldsymbol{V}_i, \boldsymbol{V}_j)}{\sqrt{\mathrm{Var}(\boldsymbol{V}_i)\mathrm{Var}(\boldsymbol{V}_j)}} \tag{6}$$

where $\boldsymbol{V}_i$ and $\boldsymbol{V}_j$ are the value of all samples at band i and j, Cov() and Var() are the covariance and variance.

With the three metrics, we propose a QI to measure the quality of a TBC

$$\mathrm{QI}_{i,j} = \mathrm{Gini}_{i,j}^{\mathrm{stand}} + (1 - \mathrm{OL}_{i,j}) + (1 - \mathrm{Cor}_{i,j}) \tag{7}$$

where $\mathrm{Gini}_{i,j}^{\mathrm{stand}} \in [0, 1]$ is the standardized $\mathrm{Gini}_{i,j}$ to make it comparable with the other two metrics. It should be noted that we expect to obtain TBCs with low correlation, i.e., TBCs with Cor around 0. Due to the widespread high positive correlation between adjacent bands of HSI, most TBCs that meet our requirements are weakly negatively correlated (a detailed explanation can be found in Fig. S1). Therefore, here we use $1 - \mathrm{Cor}_{i,j}$ to highlight TBCs with a weakly negative correlation.

D. TBC Optimization Using Band Combination Search

In order to select an appropriate subset of TBCs in an effective manner, BCS with a coarse-fine strategy [46], [47] is adopted. The process of BCS is given as follows (Fig. 4).

1) *Generating a Coarse-Scale Quality Map:* Fig. 4(a) shows a fine-scale quality map $\mathrm{QM}^f \in \mathbb{R}^{c\times c}$ for an HSI with 270 bands, where the value of every pixel is the QI of the corresponding TBC. Searching for the optimal TBCs under all possible TBCs is a very time-consuming process. Since adjacent bands are highly correlated with each other, we adopt a uniformly grouping strategy to reduce the number of TBCs that need to be computed without complicating the algorithm

$$\mathrm{Interval}_g = \begin{cases} d, & g < G \\ c - \sum_{g=1}^{G-1} \mathrm{Interval}_j, & g = G \end{cases} \tag{8}$$

where $\mathrm{Interval}_g$ means the band number in the gth band group, d is a user-defined parameter to control grouping interval, G is the number of groups and is equal to $\mathrm{Round}(c/d)$. So instead of producing the QM^f, we use the average spectrum to represent each group; thus, a coarse-scale quality map $\mathrm{QM}^c \in \mathbb{R}^{G\times G}$ can be formed [Fig. 4(b)], where the value of pixel $p_{i,j}^c$ in this map is the QI of corresponding TBC of spectral-averaged bands of the ith and jth band groups. Clearly, QM^c is a symmetric matrix, only the lower half needs to be calculated, and every pixel $p_{i,j}^c$ on QM^c has a corresponding local fine-scale quality map $\mathrm{QM}_{i,j}^{\mathrm{lf}} \in \mathbb{R}^{d\times d}$.

2) *Searching for TBCs in Local Fine-Scale Quality Maps:* We first select pixel $p_{i,j}^c$ with the highest QI on QM^c, and then find pixel $p_{k,l}^f$ with the highest QI on $\mathrm{QM}_{i,j}^{\mathrm{lf}}$. Therefore, (b_k, b_l) is the TBC we searched for with the current QM^c [Fig. 4(e)–(g). Note that only when a pixel $p_{i,j}^c$ on QM^c is selected, its corresponding $\mathrm{QM}_{i,j}^{\mathrm{lf}}$ needs to be calculated.

3) *Coarse Quality Map Masking:* Uniform grouping of bands will not completely eliminate correlations between groups. For a pixel $p_{i,j}^c$ selected in 2), we define a similarity window centered on it, where the pixels are masked and assigned a value of 0 [Fig. 4(c) and (d)]. The size of this similarity window is

$$W_s = \mathrm{Round}\left(\frac{c}{sd}\right). \tag{9}$$

By iteratively performing the operations in 2) and 3), we can obtain the specified number of TBCs. With the above process, BCS greatly reduces the computational burden in finding an optimal TBC set. Note that we specified a different number of TBCs to compare ISSFSP fairly with five state-of-the-art FS methods in the experiments section below without setting a stopping

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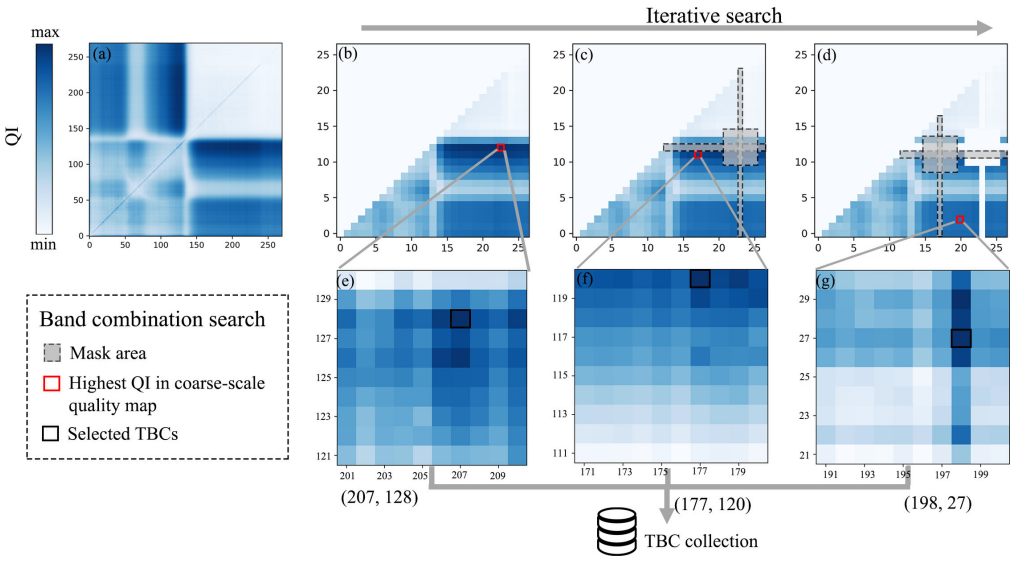


Fig. 4. BCS for SSFSP FS. (a) Fine-scale quality map. (b)–(d) 1st–3rd iterations of the coarse-scale quality map. (e)–(g) 1st–3rd iterations of the local coarse-scale quality map.

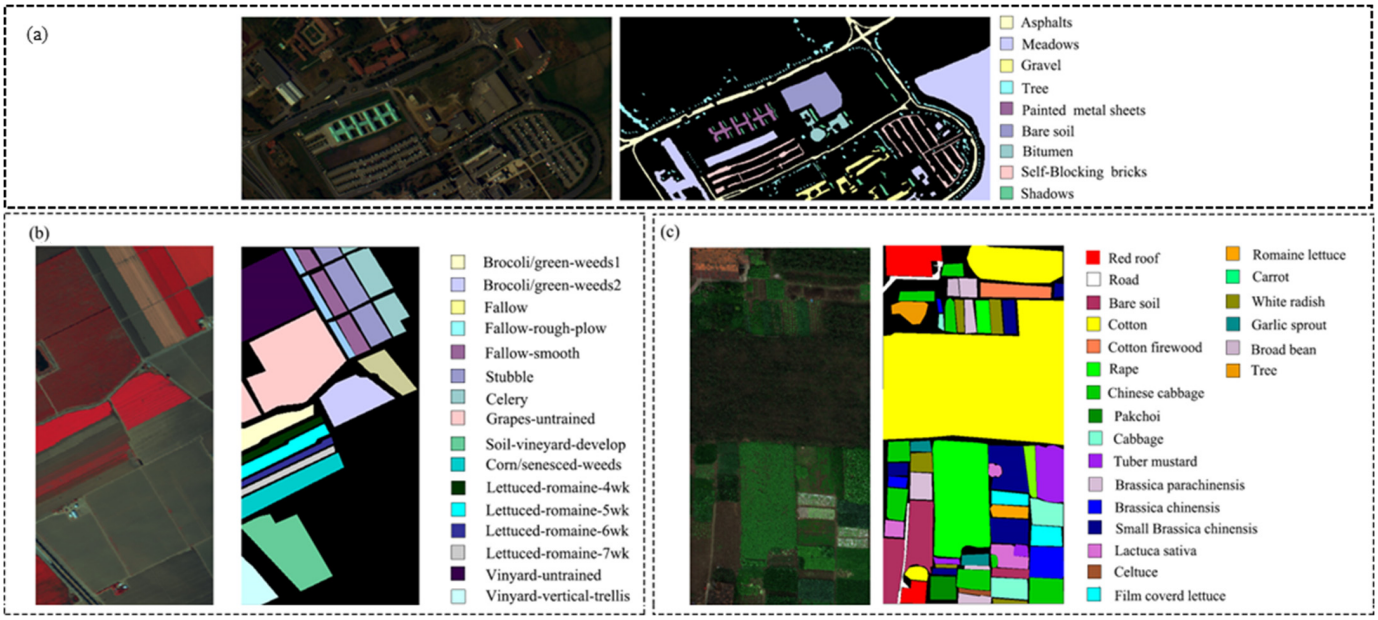


Fig. 5. False color composite images, labeled samples, and LCs information of the three datasets used. (a) PU. (b) SA. (c) WHU-Hi-HongHu (Honghu).

criterion for the iteration. In the Discussion section, we presented a method to determine the optimal number of TBCs according to the drop of QI and validated its effectiveness by checking the convergence of classification accuracy.

IV. Experiments

In this section, three datasets, five competitive features extracted by state-of-the-art FS methods, and six classification models were adopted to validate the superiority of ISSFSP through two experiments. The first experiment was designed to test the performance of ISSFSP against features selected by five other competitive FS methods under a simple CNN. The results of the first experiment can indicate the performance of these features themselves when the effects of the model were excluded. The second experiment aimed to test the effects of classification models with different structures on ISSFSP and the other five competitive features. Therefore, the results of the second experiment can demonstrate the dependence of features on the structure of the model. Note that the experiments and analyses about the parameters of ISSFSP are placed in the discussion section.

A. Datasets Description

Three publicly available HSI datasets were used. Fig. 5 illustrates the false-color composite images, the ground-truth maps, and the labeled samples of the three datasets: The first dataset

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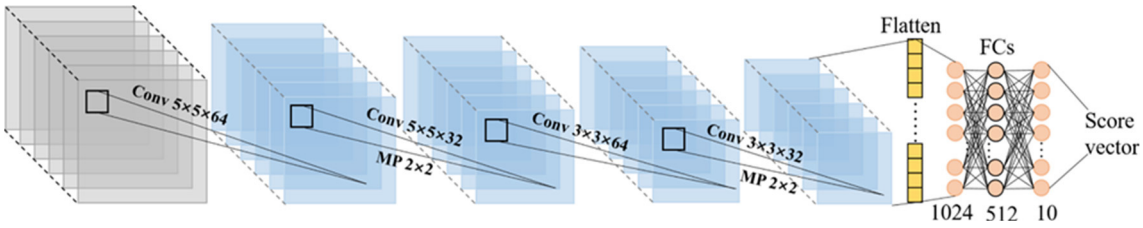


Fig. 6. Benchmark model (multi-OCNN) used in this study.

Pavia University (PU) [48] is a part of the HSI data acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor during a flight campaign over Pavia, northern Italy [Fig. 5(a)]. It contains 103 spectral bands ranging from 0.43 to 0.96 μm, covering 610 $\times$ 1096 pixels of which the spatial resolution is 1.3 m. The ground-truth map in Fig. 3 shows the 42 776 labeled pixels of nine LC types of interest. The second dataset Salinas (SA) [48] is located in a plot of farmland in the SA Valley, CA, USA [Fig. 5(b)]. The HSI was acquired by Jet Propulsion Laboratory's Airborne Visible–Infrared Imaging Spectrometer (AVIRIS) with 224 contiguous spectral bands across wavelengths from 0.4 to 2.5 μm. The image size in pixels is 512 $\times$ 217 with a high spatial resolution of 3.7 m. The ground-truth map consists of 16 LC types with 54 129 labeled samples in total. The last dataset WHU-Hi-HongHu (Honghu) [30], [49] is located in Honghu, Hubei, China. The area is a complex agricultural scene with a variety of crop types [Fig. 5(c)]. The HSI was captured by a Headwall Nano-Hyperspec imaging sensor with 270 bands ranging from 0.4 to 1.0 μm. The size of the imagery is 940 $\times$ 475 pixels with a very high spatial resolution of 0.043 m. There are 22 LC types with 386 693 labeled samples available in the ground-truth map.

B. Experimental Settings

All experiments were conducted with the Pytorch platform on Windows 10 operating system with 20 cores of Inter[1] Core[2] i9-10900K CPU at 3.70 GHz and an NVIDIA GeForce RTX 3080 graphic card with 10-GB memory.

1) *Competitors:* To verify the superiority of the proposed method, we compared ISSFSP with features extracted by five state-of-the-art FS methods (Table I).
2) *Adopted Models:* A total of six models were used for the experiments. These six models were Multi-OCNN [36] (Fig. 6), improved 3DCNN (I3DCNN) [32], two pretrained versions of the Resnet family (Resnet-18 and Resnet-80) [50], Attention-aided CNN (AACNN) [39], and a simple multilayer perceptron model (MLP). The multi-OCNN was developed for object-based classification. We adopted its main architecture, except that we used a patch input strategy to classify at the pixel level instead of the prior image segmentation. We used this model as the benchmark model. All experiments were based on this model unless otherwise stated. I3DCNN is a hybrid CNN consisting of multiple 3-D and 1-D convolutional layers. The well-known Resnet introduced the residual structure that allows the model to extract deeper features without gradient dispersion or explosion. The AACNN introduced attentional mechanisms from spectral and spatial dimensions. The MLP is not a DL model and was used to test the performance of features extracted by various methods on traditional classifiers. The topologies of the benchmark model and MLP can be found in Table S1. Without loss of generality, the logarithmic form of Softmax (Log-Softmax) was used on all models to compute the class probabilities, for it showed a better performance than Softmax [51]. The cross-entropy loss function and Adam optimizer with a batch size of 64 and weight decay of 0.0008 (other parameters of Adam referred to [52]) were used to update the parameters of the models. The training epoch for all the six models was set to 200 epochs, and the learning rate was changed to half every 40 epochs. The same training settings were adopted for all models.
3) *Parameter Settings:* There were four parameters to be set, i.e., w, R for SSFSP, and d, s for BCS. Referring to [20], w was set to 15 for the SA and Honghu datasets and 5 for the PU dataset as PU was more spatially heterogeneous, and R was set to 25 for all three datasets, indicating a GSI size of 25 $\times$ 25. d and s were empirically set to 3 and 10. In addition, considering the strong prediction capacity of the CNNs, only 5% of the labeled samples were randomly selected as the training set (also used for BCS) and the left 95% as the testing set. If not specifically stated, the values of these parameters are the same as here for ISSFSP in all subsequent experiments. The sensitivity of ISSFSP to its parameters are analyzed in the discussion section.
4) *Evaluation Metrics:* The classification accuracies in the following experiments were evaluated by three commonly used metrics, namely, the overall accuracy (OA), the average accuracy (AA), and the Cohen's Kappa coefficient (Kappa) [53], [54], [55]. OA represents the proportion of the correctly classified samples to the total samples. AA is the average of the producer's accuracies of all classes and measures how well the model can distinguish between classes. Kappa measures the classification agreement beyond the chance agreement, which is not considered by OA and is another commonly used metric in a multiclass classification task. AA and Kappa are adopted to consider the effect of sample imbalance.

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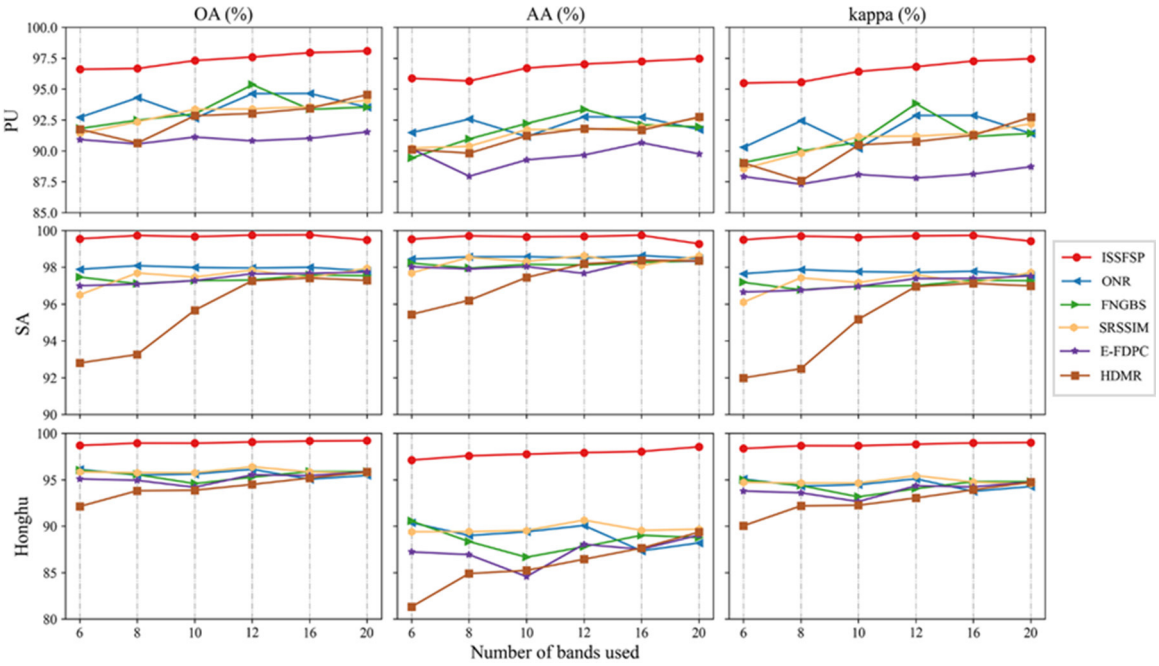


Fig. 7. OA, AA, and Kappa curves produced by combining the benchmark model with ISSFSP and competitors, respectively.

TABLE I
ADOPTED FS METHODS USED TO MAKE COMPARISONS

Category	Method
Grouping-based	Optimal Neighborhood Reconstruction (ONR) [7]
Ranking-based	Fast Neighborhood Grouping Method for Hyperspectral Band Selection (FNGBS) [9]
	Similarity-based ranking structural similarity (SR-SSIM) [8]
	Enhanced fast density-peak-based clustering (E-FDPC) [56]
Supervised	High Dimensional Model Representation (HDMR) [26]

All metrics given in the following experiments are the average of ten repetitions.

C. Experimental Results

1) *Experiment I (The Performance of ISSFPS and Competitors With the Benchmark CNN):* The purpose of the first experiment was to test the superiority of ISSFSP over the comparative FS methods under the benchmark model. To generate input patches for CNNs with the same channels, the number of bands used by ISSFSP was twice that of comparative FS methods since each GSI was generated with two bands. For a fair comparison, we tested the performance of ISSFSP and competitors with the same number of bands used. The metrics of ISSFSP and its competitors using 6–20 bands (namely, 3–10 channels for ISSFSP and 6–20 channels for competitors) were shown in Fig. 7. (The detailed statistics are given in Tables S2–S4.) As the number of channels increased, the performance of all the features generally improved. Clearly, ISSFSP consistently achieved the highest accuracy on the three datasets and had very little variation on the SA and Honghu datasets. Meanwhile, there was a clear gap between the metrics of competitors and that of ISSFSP, and some competitors were even overfitted as the number of channels increased (e.g., ONR on the PU dataset). Another interesting finding was that the differences between competitors and ISSFSP were greater on AA and Kappa than on OA (more pronounced on AA for Honghu). This was possibly because we fed the samples for each LC proportionally, and ISSFSP overcame the sample imbalance very well.

2) *Experiment II (The Performance of ISSFPS and Competitors With Various Models):* The second experiment considered the ability of classification models to extract features. If ISSFSP performs with little variation across models and all outperform competitors, then users do not need to investigate the features and models to use in a specific application. We used 12 bands for this experiment because the performance of all methods in Experiment I exhibited saturation at a band count of 12. Fig. 8 shows the OA for all methods over the three datasets under various models (the detailed statistics were placed in Tables S5–S7, and visualization of classification results were given in Figs. S2–S4), where a more yellow color indicates higher accuracy and a more green color lower accuracy. As can be seen, ISSFSP achieved the highest accuracy and did not change significantly among models. This indicated that ISSFSP was less model-dependent and outperformed competitors even with the simple MLP. In contrast, the accuracy of the competitors varied significantly under different models. Overall, all methods achieved better

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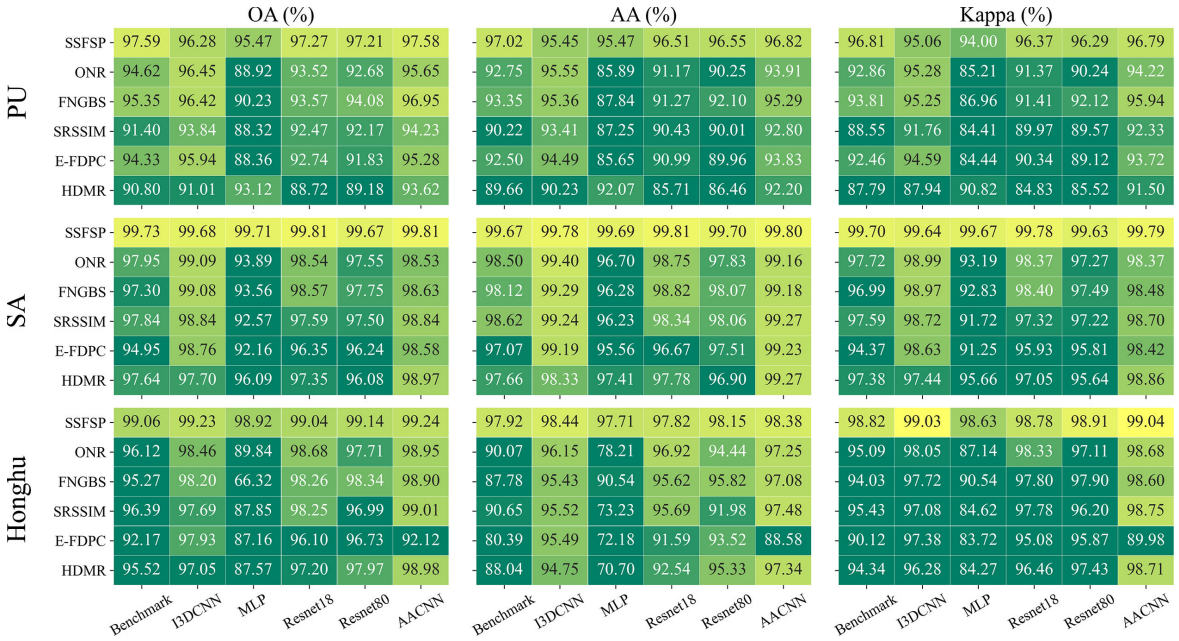


Fig. 8. OA, AA, and Kappa metrics produced by combining various CNNs with ISSFSP and competitors, respectively. The color of each grid correlates with the corresponding metric value, with higher values more yellowish and lower more greenish.

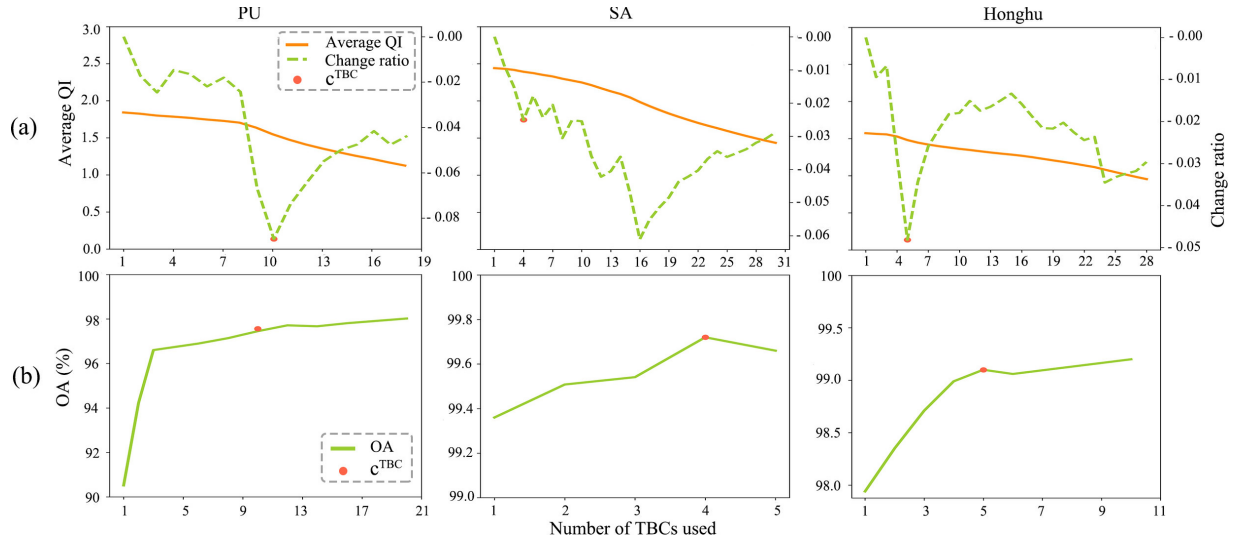


Fig. 9. Recommended number of TBCs for the three datasets. (a) Curves of AQI and CR. (b) Curves of OA.

results on both I3DCNN and AACNN, demonstrating the contribution of model improvement to the LC classification. Nevertheless, the high time consumption of AACNN due to the complex structure should not be ignored (the time consumption of various models is analyzed in the discussion section). This experiment showed the robustness of ISSFSP and its ability to achieve high accuracy without requiring a complex classification model.

V. Discussion

This section discusses some key issues related to ISSFSP involving the recommended number of TBCs, the sensitivity of ISSFSP to its parameters, the time consumption of ISSFSP at each stage, and finally, the superiority of ISSFSP. These issues are critical in terms of practical application and user guidance.

A. Recommended Number of TBC for ISSFSP

To get the recommended number of TBCs (i.e., c^{TBC}) more efficiently, we started with QI only and did not rely on any classifier. Average QI (AQI) was used to represent the average quality of the first i TBCs selected by BCS, where

$$\mathrm{AQI}_i = \frac{\sum_{k=1}^{i} \mathrm{QI}_k}{i}. \tag{10}$$

The AQI is monotonically decreasing, and its first-order derivative reflects the change rate (CR). Therefore, each local

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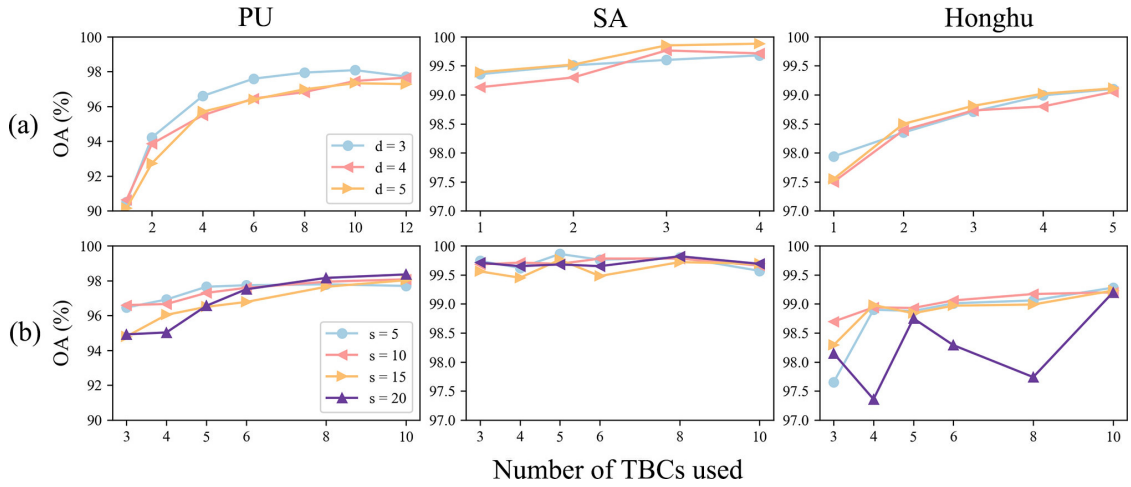


Fig. 10. Sensitivity of BCS to d and s. Row (a) is the classification OA curves with d ranging from 3 to 5. Row (b) is the classification OA curves with s ranging from 5 to 20.

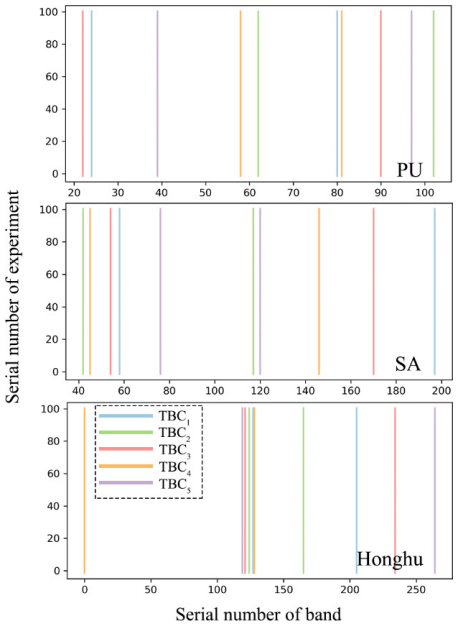


Fig. 11. Distribution of bands selected by BCS for the three datasets. Scatters of the same color represent that the corresponding bands are selected as a TBC in one experiment.

minimum on the CR curve can be treated as a stratification indicator of TBCs. In addition, for each dataset, the QI of the first TBC selected by BCS reflects the difficulty of classifying the dataset. Considering datasets that are not easily distinguishable require more TBCs, the recommended number of TBCs can be judged as

$$c^{\mathrm{TBC}} = \begin{cases} \mathrm{gm}, & \text{if } \mathrm{QI}_1 \le T \\ \mathrm{flm}, & \text{if } \mathrm{QI}_1 > T \end{cases} \tag{11}$$

where gm and flm are the number of TBCs corresponding to the global minimum and first local minimum of the CR curve, and T is a threshold to judge the difficulty of classifying a dataset. Considering the possible range of QI, T was set empirically to 2. Curves of AQI and CR of the three datasets and the corresponding classification OAs with different number of TBCs are shown in Fig. 9. The QI_1 of SA was higher than 2, therefore the recommended c^{TBC} for SA was 4 (the first local minimum). As shown in Fig. 9(b), the highest OA was achieved when four TBCs were used for SA classification, indicating the effectiveness of the recommended c^{TBC}. This strategy can also be confirmed on PU and Honghu datasets, where the QI_1 was smaller than 2, and the recommended c^{TBC} were 10 and 5 (the global minimum), respectively.

B. Sensitivity of BCS

There are four user-defined parameters for ISSFSP. As the R and w are specific to SSFSP and have already been analyzed in [20], we here analyzed the sensitivity of BCS to its parameters, namely, the scale ratio d, size of similarity window W_s, and its stability for TBC selection.

1) *Sensitivity of BCS to d:* d controls the grouping interval in (8) and was empirically set to 3 in previous experiments. Although a larger d saves more time in the TBC selection, it also causes more spectral details to be erased. Therefore, d should not be too large. The OAs of ISSFSP for the three datasets with d ranging from 3 to 5 are shown in Fig. 10(a). No significant change in OA was observed in this range. Based on the above results, we recommend a d between 3 and 5, and we still set d to 3 in the following experiments.
2) *Sensitivity of BCS to W_s:* The size of the similarity window W_s is controlled by s. Naturally, a too small s will result in too much area on QM^c being masked, while a too large s will result in redundancy between TBCs. Fig. 10(b) shows the classification OAs of ISSFSP for the three datasets with different s. As s increased, OA, in general, showed a trend of first increasing and then decreasing, so the recommended value of s for ISSFSP is 10.
3) *Stability of BCS:* The TBCs selected by BCS are expected to be stable regardless of the training samples, as it gives the user a greater level of confidence in BCS.

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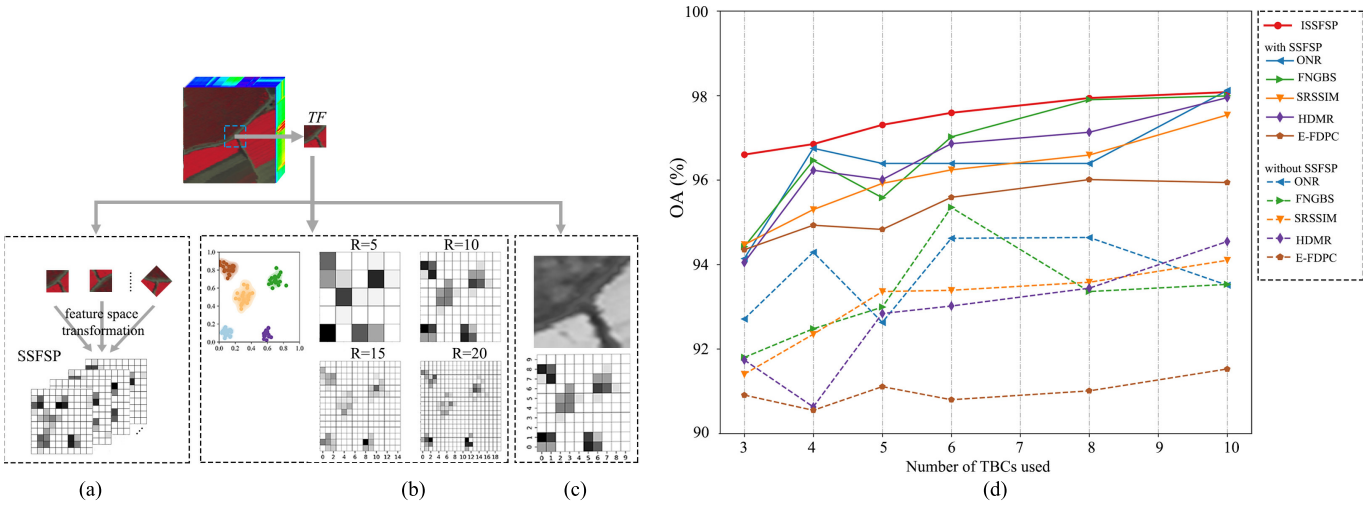


Fig. 12. Superiority of ISSFSP. (a) Feature generalization in SSFSP. (b) Feature smoothness in SSFSP. (c) Sparsity in SSFSP. (d) Classification accuracy of ISSFSP and other FS methods (without BCS and/or SSFSP). The dashed lines are the OAs of directly inputting the features to CNN, and the solid lines are the OAs of transforming the features into SSFSP and then input to CNN.

TABLE II
ABLATION EXPERIMENT OF BCS USING THE BENCHMARK MODEL AND PU DATASET

TBC selection strategy	number of TBCs					
	3	4	5	6	8	10
(a) Randomly select bands from the original bands to form *n* TBCs	73.58±0.82	78.49±0.49	79.50±0.57	80.85±0.87	82.20±1.12	85.28±0.38
(b) Randomly select bands from coarse-scale quality map to form *n* TBCs	78.77±1.72	85.43±2.06	88.35±1.21	90.87±0.91	92.38±0.31	93.66±0.56
(c) Select TBCs with top *n* QIs using the coarse-scale quality map	77.69±0.33	76.28±0.34	78.52±0.87	78.73±0.11	81.13±0.42	84.33±0.58
(d) Select TBCs with top *n* QIs using the coarse-scale quality map and similarity window mask	79.69±0.72	87.48±0.20	90.78±0.15	91.79±0.74	94.03±0.21	94.47±0.17
(e) Select TBCs with top *n* QIs using coarse-to-fine strategy and similarity window mask	96.60±0.24	96.66±0.46	97.30±0.30	97.59±0.12	97.94±0.12	98.08±0.11

* *The values in this table are OA (%) ± standard deviation of the corresponding experiment.*

We repeated the BCS 100 times for the three datasets, selecting 5% of the training samples at random per time. The ten bands that were selected in the top five TBCs in each experiment were the same, indicating the outstanding stability of BCS (Fig. 11).

C. Time Consumption

Time consumption is a crucial consideration for an algorithm to be applied in practice. We listed the time required for each stage of the proposed ISSFSP and the competitors in Table S8. Although the search space of BCS was much larger than that of competitors with the same band set, the time required by BCS was relatively acceptable. In general, since ISSFSP is transformed from TFP and usually has a larger size in spatial dimensions, ISSFSP requires more time to generate data and train the model than its competitors; however, ISSFSP has been confirmed to have low dependence on model structure (Fig. 8) and can achieve high classification accuracy with the simple benchmark model (even with MLP), thus reducing the time for model training (MT) compared to the competitors, which require more complex models. In addition, when applying ISSFSP for a large-scale region, the required time for data generation (DG) is exponentially related to the scene size in a pixel-level classification situation. Since ISSFSP has no requirement on the input TFP in terms of geometry, we can perform the segmentation operation first and then perform the object-level classification, thus saving the time spent in DG significantly.

D. Superiority of ISSFSP

The superiority of ISSFSP was analyzed from two aspects, namely, the SSFSP and BCS. The advantages of SSFSP can be summarized in three points. First, it possesses a high capability of feature generalization. For TFPs with the same pixels but different spatial arrangements, the same SSFSP was generated [Fig. 12(a)]. This characteristic allows SSFSP to reduce the requirement for sample quality. In particular, the effect of sample imbalance is significantly alleviated. Second, the feature space partitioning of SSFSP significantly reduces intraclass variance, and the SSFSP can tradeoff between

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intraclass and interclass variance when setting an appropriate value of R [Fig. 12(b)], which is conducive to improving classification accuracy. Finally, SSFSP is a sparse feature. Compared to TFP, where all pixels are involved in information extraction, only a small number of pixels of SSFSP need to be focused on to complete the FE [Fig. 12(c)] [57]. This characteristic is beneficial to the MT and application, which explains why SSFSP can even achieve good classification accuracy on MLP.

As stated in Section I, ISSFSP is the modification of SSFSP. A direct comparison of SSFSP and ISSFSP can preliminarily show the superiority of ISSFSP over SSFSP: With three TBCs, the SSFSP got 93.77% OA, 93.44% AA, and 93.06 kappa for the SA dataset [20], while the ISSFSP got 99.54% OA, 99.51% AA, and 99.49% kappa. As for the Honghu dataset, the SSFSP and ISSFSP achieved comparable accuracies. The SSFSP got 98.95% OA, 97.37% AA, and 98.67% kappa [20] and ISSFSP got 98.69% OA, 97.10% AA, and 98.34% kappa. In the following, we explored the respective contributions of SSFSP and BCS to classification accuracy improvement. We first conducted two controlled experiments with the PU dataset: (1) LC classification with SSFSP versus without SSFSP: a competitor-SSFSP-CNN set where SSFSPs were generated from bands selected by the competitors and used as the input to CNN and a competitor-CNN set where the bands selected by the competitors are directly used as the input to the CNN were compared to test the role of SSFSP in classification and (2) FS with BCS versus its competitors: BCS-SSFSP-CNN and the competitor-SSFSP-CNN set were compared to test whether BCS was the optimal method relative to the competitors. In the competitor-SSFSP-CNN set, TBCs were generated using the competitor FS method by randomly selecting two bands each time without repetition. Results are shown in Fig. 12(d). In the first control group, the OAs of the competitor-SSFSP-CNN set (solid line set) were significantly improved compared to their corresponding ones in the competitor-CNN set (dashed line set). In the second control group, the OAs of BCS-SSFSP-CNN were consistently the highest (the red solid line). The results indicate that SSFSP was effective in improving the discriminability of features selected by FS methods. Meanwhile, the combination of BCS and SSFSP was the optimal solution.

We then conducted an ablation experiment to validate the components of BCS, and the settings and results are listed in Table II. Comparing (a) and (b) shows that simple group averaging can partially eliminate the correlation of bands in the original dataset, but random selection increased the instability of the results (see the corresponding standard deviations). In (c), only QI was used to select bands, and its classification accuracies were lower than that of (b). (d) added similarity window mask to (c) and got a significant improvement in both accuracy and stability compared to (c), (b), and (a). Finally, with the addition of the coarse-to-fine strategy, the classification accuracies of (e) were significantly improved, and all ranked first. In summary, group averaging can be used as an effective method to reduce computation. If, however, only QI is used to select bands without using the similarity window mask, the selected TBCs with high QIs will have increased correlation because these TBCs are concentrated on the coarse-scale quality map. Therefore, QI and similarity window mask need to be used together to find quality TBCs. Moreover, the coarse-to-fine strategy helps to find finer spectral features after the appropriate band ranges are found by the former operations.

VI. Conclusion

In this article, we proposed ISSFSP for CNN-based HSI classification by adding a novel FS method, BCS, to the SSFSP method. We validated the performance of ISSFSP by experiments involving three HSI datasets, five other state-of-the-art FS methods, and six classification models. Results indicate that ISSFSP not only achieves the best classification accuracy but also reduces the reliance on model structure and sample quality. The superiority of ISSFSP can be attributed to the fact that ISSFSP is an advanced spectral representation that transforms the implicit spectral relationships to be processed explicitly by CNN. Meanwhile, BCS can automatically select an optimal set of TBCs for the target LCs and determine the required number of TBCs to produce smaller data while obtaining better classification accuracy. Since ISSFSP improves classification accuracy while significantly reducing the reliance on expertise and user interaction, we believe that it has great potential for practical use and is worth promoting.

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